

Circuit Analysis (Lab10)

Experiment 9: Verification of Thevenin's Theorem

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Grade and Signature:

CLO2 CLO3	Use with identification the circuit components, breadboards, multimeter, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)				
	Total marks (10)			

Objective

The objective of this lab is verify Thevenin's theorem and to find the full load current.

Apparatus Required

Sr. no	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	1k, 330 ohms	1,3
7	Potentiometer	Standard	1

Description

Statement:

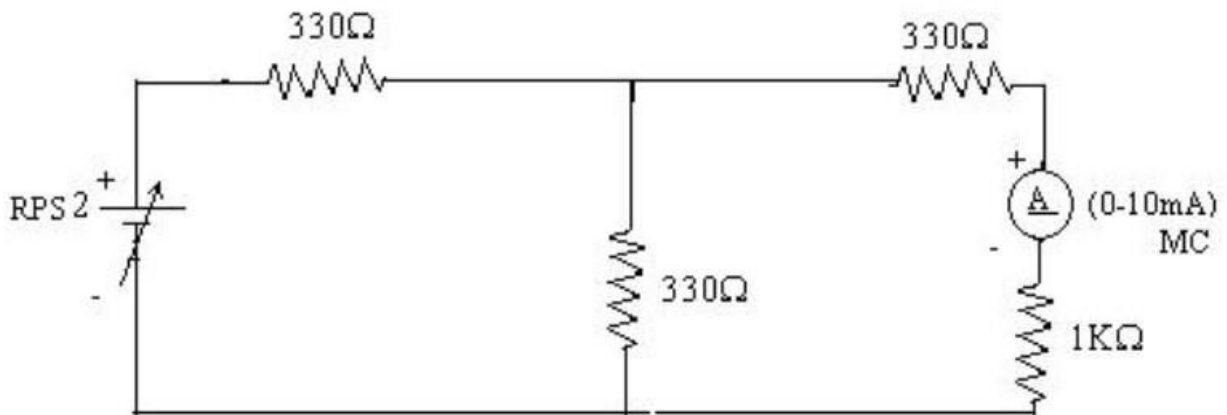
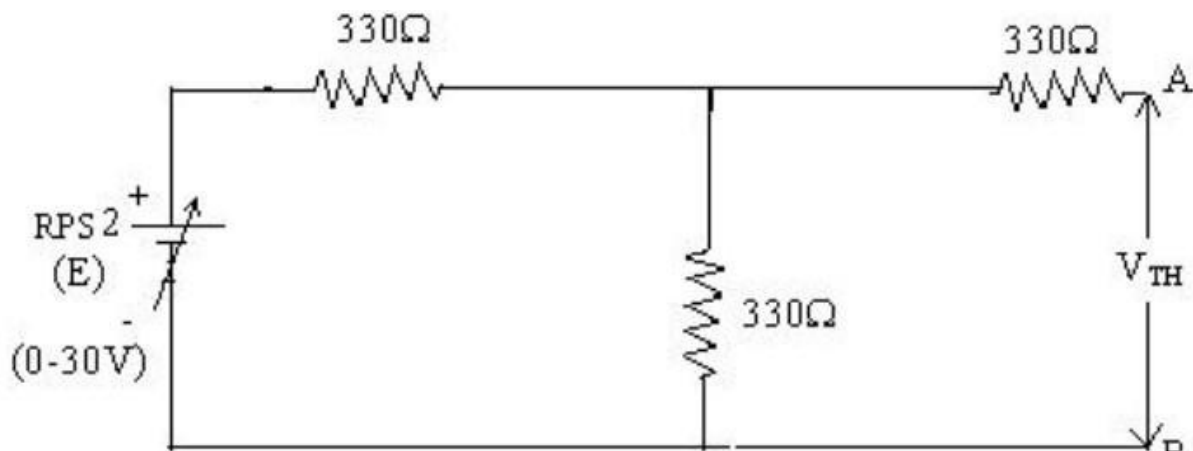
Any linear bilateral, active two terminal network can be replaced by an equivalent voltage source (VTH). Thevenin's voltage or V OC in series with looking back resistance RTH.

Precautions:

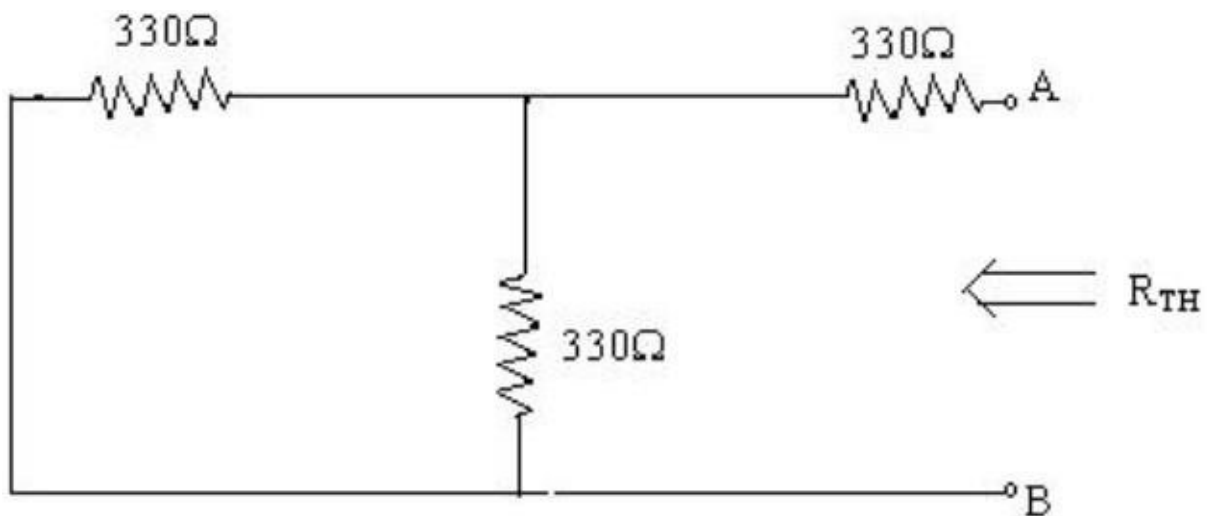
1. Voltage control knob of RPS should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position

Procedure:

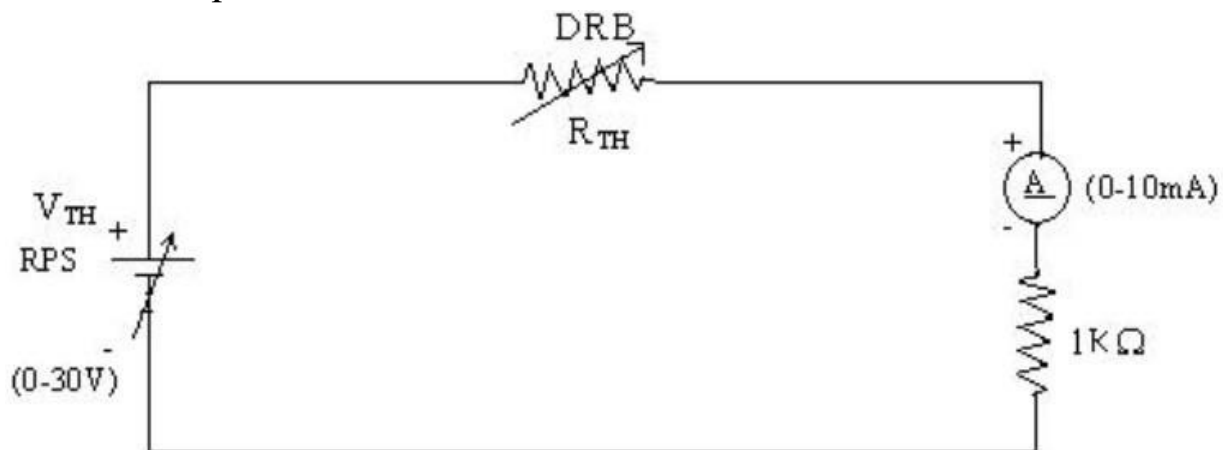
1. Connections are given as per the circuit diagram.
2. Set a particular value of voltage using RPS and note down the corresponding ammeter readings. To find V_{TH}
3. Remove the load resistance and measure the open circuit voltage using multimeter (V_{TH}). To find R_{TH}
4. To find the Thevenin's resistance, remove the RPS and short circuit it and find the R_{TH} using multimeter.
5. Give the connections for equivalent circuit and set V_{TH} and R_{TH} and note the corresponding ammeter reading.
6. Verify Thevenin's theorem.

Circuit - 1: To find load current**To find V_{TH}** 

To find R_{TH}



Thevenin's Equivalent circuit:



Thevenin Practical Values

	E(V)	$V_{TH}(V)$	$R_{TH}(\text{ohms})$	$I_L \text{ (mA)}$	
				Circuit - I	Equivalent Circuit
Values	10V	5V	495	3.34	3.34

Post Lab Questions

1. What is Thevenin's Theorem?

A method used to simplify a complex linear circuit into a much simpler equivalent version. It states that any combination of voltage sources, current sources, and resistors with two output terminals can be replaced by a single voltage source called V_{th} (Thevenin Voltage) connected in series with a single resistor called R_{th} (Thevenin Resistance). This makes it much easier to calculate what happens when you change the load connected to those terminals.

2. What are the applications of Thevenin's Theorem?

1. Circuit Simplification: It helps in reducing large, complicated networks into a manageable form for faster analysis.
2. Power Systems: Engineers use it to analyze power grids and determine how much current will flow during a fault or when a new load is added.
3. Varying Load Analysis: It is perfect for situations where the "source" part of the circuit stays the same but the "load" (like a speaker or a motor) changes frequently.
4. Battery Modeling: It is used to represent real-world batteries, where the V_{th} is the ideal battery voltage and R_{th} represents the internal resistance of the battery.

3. What are the theoretical and practical limitations of Thevenin's Theorem?

Theoretical Limitations:

1. Linearity Requirement: The theorem only works for linear circuits. If a circuit contains non-linear components like transistors or diodes (where the resistance changes with voltage), it cannot be simplified using standard Thevenin steps.
2. Independent Sources: It is mainly designed for circuits with independent sources. If a circuit has dependent sources (sources that change based on another part of the circuit), the calculation for R_{th} becomes much more complex.
3. Time-Invariant Systems: It assumes the components do not change their values over time.

Practical Limitations:

1. Internal Resistance: In a real lab, measuring V_{th} and R_{th} can be slightly off because the wires and the internal components of the power supply have their own small resistances that aren't usually written on the diagram.
2. Power Loss: While the equivalent circuit gives the correct voltage and current at the load, it does not always show the correct power dissipation or efficiency of the original complex circuit.
3. Component Tolerances: Real resistors have a tolerance (like plus/minus 5%), so the practical R_{th} measured on a multimeter might differ from the value calculated on paper.

Steps (for Proteus) and Explanation of Circuits

1. Original Circuit (To Find Load Current)

This circuit measures the baseline current through the 1k load resistor before any changes are made. The current flows from the 10V source through the resistor network, splitting at the junction before reaching the ammeter and load.

Proteus Steps:

- In proteus selected the dc battery, four resistors a ground terminal and a dc ammeter.
- Constructed the circuit with the 10V battery and the resistor network placing the 1k load at the far right.
- Connected the ammeter in series with the load and set it to milliamps.

2. Circuit for Finding Thevenin Voltage

We find V_{th} by unplugging the load resistor and ammeter creating an open circuit. Since the path is broken no current flows through the final resistor and the voltmeter across the open points reads the Thevenin voltage.

Proteus Steps:

- Removed the 1k load and ammeter from the circuit.
- Selected a dc voltmeter and connected it across the open output terminals.
- Ran the simulation to record the open circuit voltage.

3. Circuit for Finding Thevenin Resistance

To get R_{th} we turn off the power source and replace it with a wire (short circuit). Looking back from the open output, we measure the total internal resistance the load would normally see

Proteus Steps:

- Replaced the battery with a wire to ground to short the input.
- Selected the ohmmeter component and connected it to the output terminals.
- Noted the resistance value shown on the meter.

4. Thevenin Equivalent Circuit

We replace the whole network with one battery (V_{th}) and one resistor (R_{th}) in series. Adding the original 1k load back shows the exact same current proving the theorem works.

Proteus Steps:

- Built a new loop with a battery set to V_{th} and a resistor set to R_{th} .
- Reconnected the original 1k load and ammeter in series.
- Verified the current reading matched the original circuit.

Circuits

